

Ziyue Fang

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EDUCATION

Columbia University

New York City, NY

B.S. in Computer Science, GPA: 3.6/4.0

Expected May 2028

- **Relevant Courses:** Artificial Intelligence, Machine Learning, Advanced Programming, Data Structures, Discrete Mathematics, Linear Algebra

EXPERIENCE

Claim

New York City, NY

Product Engineer, Internship

Mar. 2026 – Sept. 2026

- Designed an event-driven scanning pipeline on the Gmail API that matches users' purchases to class-action settlements they can claim, handling 100K+ emails a day by syncing incrementally to avoid repeated work, with encrypted token storage.
- Added parallel scanning that processes a 3-year inbox in under 10 seconds, balancing accuracy, speed, and cost with LLM verification at ~\$0.01 a scan.
- Worked with PMs and designers to A/B test the paywall on a Firebase + React (TypeScript) app, improving conversion and retention.

Zoomob Limited

Hong Kong, China

Software Engineer, Internship

May 2025 – July 2025

- Developed a full-stack AI web app (JavaScript, React, TailwindCSS) that turns user-built worlds into storylines, including the world-building UI, the generation flow, and 30+ reusable components end to end.
- Leveraged 5+ third-party API services (Google OAuth, Stripe Checkout, OpenRouter) and ran subscriptions over webhooks, from login through payment to AI generation.

PROJECTS

CollegePilot | *TypeScript, React, Supabase Edge Functions, Postgres/pgvector, RAG, LLMs*

- Built and independently scaled CollegePilot, an AI college-counseling app, to 5K+ daily active users.
- Built a RAG layer that fetches only the relevant pieces of a student's essays, activities, and the colleges database (Postgres/pgvector, 384-dim embeddings, cosine search) instead of pasting everything into the prompt, for sharper grounded feedback at far lower token cost.
- Ran A/B tests on UI/UX, onboarding, and funnels to improve activation and retention.

StonksGPT | *Node.js, Express.js, React, TypeScript, MCP, LLMs, Docker*

- Built an LLM trading agent on the Alpaca API that takes a plain-English request and runs research, manages the portfolio, or places trades through an MCP tool-calling loop, with the Node backend and Python MCP server containerized in Docker to bundle both into one reproducible image.
- Cut token costs by roughly 90% by implementing a sliding context window and LLM summarization layer, compressing each MCP tool result to just the fields the model needed.

MovieMatch | *TypeScript, React, Next.js, Python, FastAPI, Sentence Transformers, Hugging Face*

- Developed a Tinder-style movie recommender over 15K movies, ranking by cosine-similarity KNN on Hugging Face SBERT embeddings against the average of a user's likes, with a per-genre cap for variety.
- Served results from a FastAPI REST backend to a React (Next.js, TypeScript) frontend, with client-side state to cut network requests.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Express, FastAPI, Socket.IO, Tailwind CSS

Cloud & Databases: AWS (S3), GCP (Firebase), PostgreSQL, Redis

APIs & Auth: REST APIs, WebSockets, OAuth, session auth, Stripe, Webhooks, Google APIs

Tools & Practices: Git, Docker, CI/CD, event-driven architecture, A/B testing

AI/ML: LLMs, RAG, MCP, embeddings & vector search, PyTorch